VP09 — Dataset, Datagram and persistent Database

Revision: `799b274252245be2ac5258510067da66a08bb45276538bcbb6482b3189e898c3`.

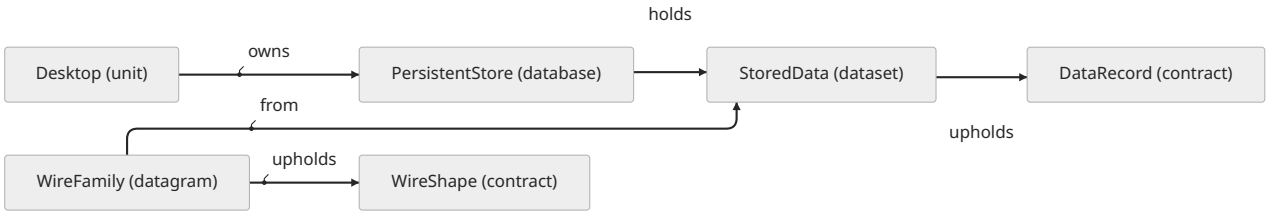
Explicit holders, sources, contracts, variants, fields and projections.

Data origin and holder: MemoryData

Source facts: f0027, f0045.

Declarations: DataRecord, Desktop, MemoryData.

Data origin and holder: StoredData



Source facts: f0029, f0067, f0074, f0076, f0077.

Declarations: DataRecord, Desktop, PersistentStore, StoredData, WireFamily, WireShape.

Scope and notation

Dataset neighborhoods retain explicit from, holds, owns and upholds relations. Database means a persistent data source, not SQL tables, keys, transactions, a storage engine or deployment. Unit-held data has no declared persistence. These are definitions, not runtime instances. Contract completeness is a source assertion, not

execution evidence. Optional fields do not imply null, empty, false or zero. Projections do not specify executable transformations. Encoding and bit positions belong to VP10.

Data definitions

- MemoryData (Dataset): holder Desktop; contract DataRecord; Unit-held; persistence unspecified. Source facts: f0027, f0045.
- StoredData (Dataset): holder PersistentStore; contract DataRecord; Database-backed persistence. Source facts: f0067, f0074.
- WireFamily (Datagram): logical source Dataset StoredData; contract WireShape. This does not inherit the source's holder or persistence. Source facts: f0076, f0077 .

Database register

Database	Responsible owner	Held Datasets	Source facts
PersistentStore	Desktop	StoredData	f0029, f0067
UnusedStore	Desktop	No Datasets declared	f0030

Contract register

All Contracts are included, even unused/open definitions. Open a Contract in the navigation tree for its structure diagram. Counts describe explicit shape membership (fields / variants / permitted messages), not runtime coverage.

Contract	Completeness	Fields / variants / messages	Source fact
ChannelBody	closed	1 / 0 / 0	f0009
ChannelEvents	closed	0 / 0 / 2	f0011
DataRecord	closed	1 / 0 / 0	f0020
UnusedShape	open	0 / 0 / 0	f0075
WireShape	closed	1 / 2 / 0	f0080

Variant register

Variant	Contract	Source fact
PacketBody	WireShape	f0078
WireVariant	WireShape	f0079

Field register

Field	Shape owner	Value type	Presence	Source facts
ChannelValue	ChannelBody	text	required	f0010, f0018, f0019
DataValue	DataRecord	text	required	f0021, f0022, f0023
PacketFlag	WireShape	boolean	required	f0055, f0056, f0081
PacketNumber	PacketBody	signed	required	f0053, f0063, f0064
PacketPayload	PacketBody	bytes	required	f0054, f0065, f0066
WireValue	WireVariant	bytes	optional	f0082, f0083, f0084

Shape membership

Shape	Relation	Member	Source fact
ChannelBody	has-field	ChannelValue	f0010
ChannelEvents	permits	ChannelNotice	f0012
ChannelEvents	permits	ChannelUpdate	f0013
DataRecord	has-field	DataValue	f0021
PacketBody	has-field	PacketNumber	f0053
PacketBody	has-field	PacketPayload	f0054
WireShape	defines	PacketBody	f0078

WireShape	defines	WireVariant	f0079
WireShape	has-field	PacketFlag	f0081
WireVariant	has-field	WireValue	f0084

Projection responsibilities

Projects relates Functionality, Dataset and Datagram together; it does not imply a lossless conversion or a runtime call.

Functionality	Dataset	Datagram	Responsible owner	Source facts
Display	StoredData	WireFamily	Desktop	f0028, f0038